

K-12 AI Whiteboard Master Toolkit

Exhaustive Subject & Grade-Wise Interactive Tools & Architectural Blueprint (v2.0)

- FULLY EXHAUSTIVE SPEC
- GEOGEBRA SUITE NATIVE
- K-12 ALL SUBJECTS
- CLOUD JSON PIPELINE
- SOCRATIC AI ENGINE

Architectural Scope & Enhancement Overview (v2.0)

This updated blueprint expands the toolkits into an **exhaustive, subject-by-subject and grade-band-categorized matrix**. It replaces generic graphing widgets with the complete **GeoGebra Engine Suite** (Graphing, Geometry, 3D, CAS) and adds dedicated toolkits for **Mathematics, Natural Sciences, Language Arts, World Languages, Social Studies/Economics, Computer Science, and Special Accessibility Needs**.

1. CORE CANVAS INFRASTRUCTURE & GEOGEBRA INTEGRATION

1.1 GeoGebra Engine Integration & Vector Canvas

GEOGEBRA NATIVE

- **Native GeoGebra Suite Injection:** Embedded GeoGebra API supporting seamless 2D Graphing, Dynamic Geometry Constructions, 3D Surface/Vector plotting, and Computer Algebra System (CAS) symbolic solving directly on the whiteboard canvas.
- **Vector-Native Hybrid Engine (WebGL + SVG):** Infinite zoom (10% to 5000%) with zero pixelation or line degradation. Built for stylus pressure sensitivity, touch gestures, and high-precision mouse operations.
- **Dual-Layer Canvas Architecture:**
 - *Active Student Workspace:* Interactive layer containing student drawing strokes, live manipulatives, and editable problem blocks.
 - *Private Tutor Overlay:* Semi-transparent control layer visible ONLY to the tutor containing step-by-step mark schemes, answer keys, timing benchmarks, and pedagogical tips.
- **Multi-User Real-Time Sync:** WebSockets/WebRTC RTC synchronization supporting concurrent multi-user pen inputs with sub-30ms latency, custom cursor ghosting, and live selection bounding boxes.

1.2 Cloud JSON Question Bank & Vector Rendering Engine

CLOUD ARCHITECTURE

PIPELINE STAGE	TECHNICAL FUNCTION	PRACTICAL BENEFIT
Cloud JSON Streaming	Side-dock query pulls indexed JSON containing stem LaTeX, diagram SVG paths, and solution trees.	Bandwidth usage < 5KB per item. Instant load under 50ms.
Client Vector Parsing	KaTeX parses LaTeX into SVG paths; diagrams render as un-groupable SVG elements on the board.	Tutors/students can isolate, rotate, recolor, or measure individual diagram components.
Dynamic Re-Rolling	"Generate Similar" triggers variable recalculation in the cloud, re-rendering math, solutions, and graphics on the fly.	Endless targeted drill practice without changing board layouts.

2. EXHAUSTIVE SUBJECT & GRADE-WISE TOOLKITS

2.1 Mathematics Toolkit (Grades K–12)

GRADE BAND	DEDICATED TOOLKITS & MANIPULATIVES	INTERACTIVE MECHANICS & FUNCTIONAL FEATURES
K–2 (Early Math)	Base-10 Blocks Ten-Frames Counting Bears Rekenrek Abacus Number Line (0-20) Analog Clock Virtual Currency Pattern Blocks	Drag-and-drop units, rods, and flats with snap-and-break animations. Ten-frames auto-count counters with subitizing audio cues. Rekenrek abacus frame with sliding beads. Analog clock with linked gear movement updates digital time. Virtual coins/bills with auto-totaling counting mats.
3–5 (Elementary)	Fraction Strips & Towers Cuisenaire Rods Geoboard Place Value Disks Multiplication Array Interactive Protractor Balance Scale	Fraction strips split dynamically (1–12) with shade, compare, and equivalence overlay toggles. Geoboard with elastic band stretching for perimeter/area calculation. Array builder visualizes partial products. Place value disks support operations up to millions and decimals.
6–8 (Middle School)	Ratio Double Lines Tape Diagrams Algebra Tiles Coordinate Grid Engine Probability Spinners/Dice Integer Chips 3D Polyhedra Nets	Algebra tiles with positive/negative x , x^2 , 1 pieces and zero-pair cancellation animations. Coordinate plane snaps to grid with auto-labeling (x,y) , slope calculation, and distance formula. Probability sandbox runs 1000-spin simulations with theoretical vs empirical graphs. Unfolding 3D nets animate into closed 3D solids.
9–12 (Advanced High School)	GeoGebra Graphing GeoGebra Geometry GeoGebra 3D Engine GeoGebra CAS Trig Unit Circle Matrix Grid Stat Distribution Plotter	Direct GeoGebra canvas embedding: plot functions, polar curves, inequalities, locus lines, conic sections, 3D vectors/surfaces, and CAS symbolic solving. Unit circle visualizer syncs angle rotation with real-time sine/cosine/tangent wave rendering. Statistical tool plots normal/binomial curves with z-score area shading.

2.2 Natural Sciences Toolkit (Physics, Chemistry, Biology & Earth)

Physics Sandbox (Grades 6–12) PHYSICS Force Vector & Free Body Diagram (FBD) Engine: Draw forces on objects; system automatically resolves F_x, F_y components, net force vector, normal force, and friction on inclined planes. Kinematics & Projectile Motion Launchpad: Adjust velocity, launch angle, height, and gravity (g) to plot real-time parabolic trajectories with air resistance toggles. Schematic Circuit Canvas: Drag-and-drop resistors, capacitors, inductors, AC/DC sources, switches, ammeters, and voltmeters. Circuit solver simulates real-time current flow, voltage drops, and oscilloscope waveforms. Ray Optics Bench: Lenses (concave/convex), plane/curved mirrors, and glass prisms. Light rays Snell's Law refract and reflect automatically based on refractive indices and focal lengths. Wave Interference & Doppler Simulator: Ripple tank and sound wave visualizers showing constructive/destructive interference, diffraction, and frequency shifts.	
Chemistry Workbench (Grades 6–12) CHEMISTRY Interactive Dynamic Periodic Table: Filter by blocks, oxidation states, or electronegativity heatmaps. Hovering provides full electron configuration shell animations; dragging elements creates atomic badges. 2D Lewis Structure Drawer & 3D VSEPR Rotator: Draw single, double, and triple bonds with lone-pair dot snaps. One-click toggle converts 2D drawings into 3D rotatable molecular geometries. Chemical Equation Balancer & Stoichiometry Engine: Auto-checking coefficient fields for reactants/products with limiting reagent and yield calculators. Gas Laws Particle Simulator: Container particle motion sandbox dynamically linked to $PV = nRT$ variables (Pressure, Volume, Temperature sliders). Titration & pH Buffer Plotter: Burette simulation plotting real-time pH titration curves for strong/weak acid-base combinations.	

Biology, Ecology & Earth Science (Grades K–12)

BIOLOGY & EARTH

- **Multi-Layer SVG Anatomy & Cell Stamps:** Cell organelle, plant tissue, and human body system overlays with "Quiz Mode" label masking.
- **Punnett Square Matrix & Pedigree Chart Builder:** \$2 imes2\$ and \$4 imes4\$ genetics grids with auto-calculated genotype/phenotype ratios and pedigree inheritance mapping.
- **Codon Ring & Protein Synthesis Tool:** Interactive mRNA codon wheel converting DNA/RNA sequences into amino acid polypeptide chains.
- **Topographic & Plate Tectonics Sandbox:** Contour line drawer, fault line earthquake simulators, and carbon/water cycle process flow blocks.

2.3 ELA, Literacy & Language Arts Toolkits

English Language Arts (ELA K–12)

LITERACY & WRITING

- **Phonics Sound Tiles & Elkonin Boxes (K–3):** Color-coded tiles for consonants, vowels, digraphs, and blends paired with segmenting boxes and audio phonemes.
- **Decodable Reader Frame (K–5):** Line-focus reading mask, OpenDyslexic font toggle, syllable splitters, and text-to-speech tracking.
- **Interactive Sentence Diagrammer (6–8):** Reed-Kellogg syntax tree builder for visually parsing subjects, verbs, direct objects, and subordinate clauses.
- **Writing Architecture Blocks (9–12):** Locking structural containers for CER (Claim, Evidence, Reasoning) and PEEL (Point, Evidence, Explanation, Link) essay composition.
- **Rhetorical Analysis & Poetry Scansion Tool:** Color-coded highlighter palette for ethos/pathos/logos and iambic pentameter scansion notation.

2.4 Foreign / World Languages (EFL, ESL, Spanish, French, German, Mandarin, etc.)

World Languages & Second-Language Acquisition (K–12)

WORLD LANGUAGES

- **Floating Diacritic & Script Overlay Keyboards:** Instant canvas access to language characters ($\tilde{n}$, $\acute{e}$, $\ddot{u}$, ç , β) and non-Latin overlays (Pinyin, Cyrillic, Arabic, Kana).
- **Verb Conjugation Matrix Wheel:** Dynamic tables auto-populating tenses (Present, Preterite, Imperfect, Subjunctive) with missing-cell quiz modes.
- **East Asian Character Tracing Grid (Tianzige / Genkouyoushi):** Character writing grids featuring stroke-order animations and vector ink-smoothing.
- **Dialogue Bubble Audio Recorder & IPA Dock:** Roleplay speech bubbles with micro-audio clip recording, IPA phonetic notation, and dual-track translation strips.

2.5 Social Studies, Economics, CS & Special Needs

Social Studies & Economics

HISTORY & ECON

- **Multi-Era Historical Map Overlay:** Interactive map boundaries, trade routes, and migration vectors.
- **Econ Supply/Demand Grapher:** Shift sliders calculating equilibrium price/quantity, consumer surplus, and price ceilings.
- **DBQ Document Matrix:** Split-screen primary source analysis quadrant.

Computer Science (CS K–12)

CS & TECH

- **Scratch-Style Block Coding Canvas:** Drag-and-drop logic blocks for K-8 programming logic.
- **Flowchart Logic Palette:** Standard flowchart symbols with snap connectors.
- **Embedded Python/Java Console:** Syntax highlighting code runner with variable execution state viewer.

Accessibility & Neurodivergent Support Suite (K–12)

ACCESSIBILITY

- Dyslexia & Visual Processing Overlays:** High-contrast line masks, OpenDyslexic font toggles, and customizable canvas background tinting.
- Color-Blindness Palette Switcher:** Protanopia, Deuteranopia, and Tritanopia compliant color modes across all scientific/math diagrams.
- Screen Reader & Voice Dictation Engine:** Full ARIA canvas navigation accessibility and real-time speech-to-text writing assistance.

3. CLOUD QUESTION BANK & DUAL-TAXONOMY ARCHITECTURE

3.1 Dual-Taxonomy Categorization & Regional Standards

QUESTION BANK UX

CURRICULUM SYSTEM	TAXONOMY BREAKDOWN	KEY FRAMEWORKS INDEXED
US Curriculum	Grade Level $\rightarrow$ Standard System $\rightarrow$ Subject Domain $\rightarrow$ Cluster $\rightarrow$ Skill	Common Core (CCSS), NGSS, AP, SAT/ACT Prep.
UK Curriculum	Key Stage (KS1–5) $\rightarrow$ Exam Board $\rightarrow$ Subject $\rightarrow$ Tier (Higher/Foundation)	AQA, Edexcel, OCR, CIE iGCSE & A-Levels.
European / IB System	Programme (PYP / MYP / DP) $\rightarrow$ Subject Group $\rightarrow$ Level (SL / HL)	IB Diploma, European Baccalaureate, National Curricula.
General / Universal View	Curriculum-Agnostic Skill Graph (e.g., Math $\rightarrow$ Algebra $\rightarrow$ Quadratic Equations)	Cross-maps questions globally across all jurisdictions.

4. NEXT-GENERATION SOCRATIC AI ENGINE

4.1 Socratic Co-Pilot & Generative Canvas Capabilities

AI FEATURES

- Dynamic Ink-to-Simulation Engine:** Hand-drawn rough sketches (e.g., an inclined plane with a block, ray of light into a lens, or circuit loop) convert instantly into interactive, physics-simulated objects.
- Multimodal Spatial & Voice Grounding:** Recognizes speech and gestures simultaneously (e.g., saying "Calculate the surface area of this shape" while circling an object isolates the vector geometry and generates the calculation workspace).
- Discreet Socratic Tutor Sidekick:** Silent inline monitoring providing tutors with real-time whispers in their private dock (e.g., "Student made a sign error on step 2—suggest showing a number line").
- Automatic Misconception Radar:** Identifies underlying conceptual misunderstandings in student ink and generates a 30-second visual remediation card.
- Prompt-to-Manipulative Generator:** Speak or type commands (e.g., "Show algebra tiles for $(x+2)(x+3)$ ") to immediately instantiate native canvas manipulatives.
- Adaptive Scaffolding Dial:** Slider adjusts text or problem complexity on the fly from Grade 3 reading levels up to AP/College level.

5. SESSION CONTROL & ASYNCHRONOUS CONTINUITY

5.1 Session Controls

HOST TOOLS

- "Bring All to Me":** Snaps student viewports to the tutor's focal point.
- Pen Freeze & Spotlight:** Locks student drawing tools during critical explanations.
- Private Tutor Scratchpad:** Private dock for answer keys and lesson timing notes.

5.2 Post-Session Magic

ASYNCH AI

- Living Interactive Audio Notes:** PDF export embeds audio bookmarks linking handwritten lines to exact 10-second lesson voice clips.
- AI Spaced-Repetition Warm-ups:** Auto-generates a 3-minute pre-session practice board based on past friction points.